DATABASES

SQL vs NoSQL Visual Guide

A decision framework --
map your workload to the right store.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Know Your Workload

Four axes decide the answer before any product name appears.

Structure

fixed schema + relations, or flexible/evolving shapes?

Scale

GBs on one node, or TBs across many?

Consistency

must every read see the latest write?

Query patterns

ad-hoc joins, or known key lookups?

STEP 2/6

Choose SQL When...

Relational strengths, honestly stated.

Relationships are the product

orders JOIN customers JOIN items -- joins are the query

ACID is non-negotiable

money movement, inventory -- no partial writes

Queries are ad-hoc

analysts ask questions you didn't pre-model

One operational brain

your team knows Postgres -- boring is reliable

STEP 3/6

Choose NoSQL When...

Scale and shape flexibility, honestly stated.

Write scale exceeds one node

millions of writes/sec -- shard by design

Schema evolves constantly

each document carries its own shape

Access patterns are known

key lookups you can enumerate up front

Special structure fits

graphs, time series, vectors -- native models win

STEP 4/6

Consistency vs Availability

The trade you are actually making.

CP

consistent +■partition-tolerant

AP

available +■partition-tolerant

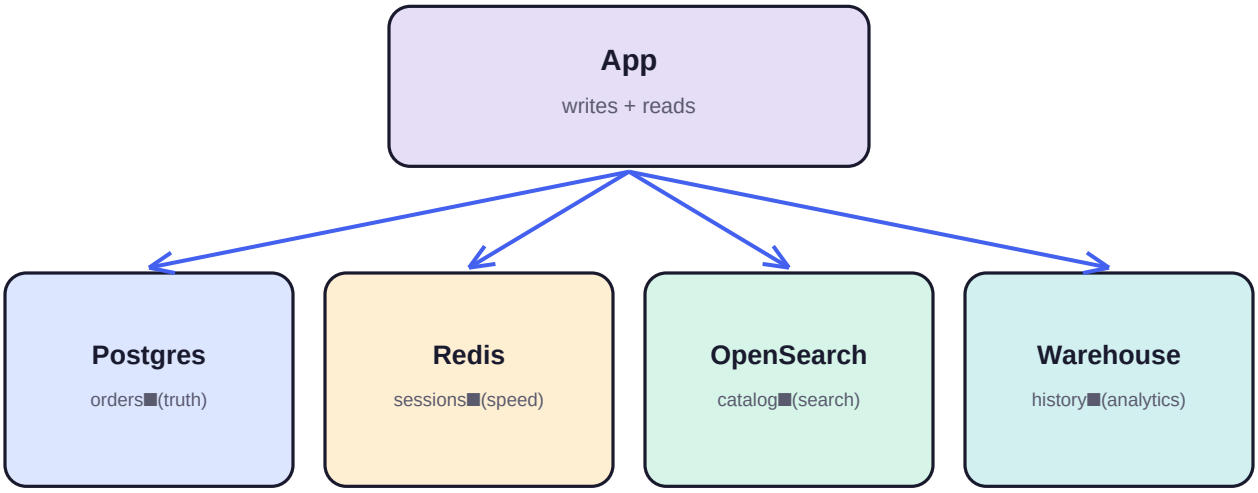
- During a network partition you pick: refuse writes (CP) or accept divergence (AP).
- Most systems live in the middle: tunable consistency (Cassandra), sync/async replicas.
- Ask: 'what breaks if two users see different data for 5 seconds?' -- that is your answer.

CAP is about partitions. Day-to-day, the real dial is latency vs consistency -- PACELC.

STEP 5/6

Polyglot Persistence

Real systems use several stores -- each for its strength.



Polyglot is powerful and expensive: every store is a backup, monitoring, and expertise bill.

STEP ✓I✓

Key Takeaways

SQL vs NoSQL -- Remember

- Decide on workload axes first: structure, scale, consistency needs, query patterns.
- SQL wins on relationships, ACID guarantees, ad-hoc queries, and operational familiarity.
- NoSQL wins on write scale, flexible schemas, known access patterns, and native data models.
- Consistency vs availability is a per-partition trade -- most systems tune the middle.
- Polyglot persistence is normal at scale -- but every added store is an operational bill.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev